Java Programming – LAB 3

1. Create a class **Student** with **rollno**, **name**, **age** and **totalmark** as instance variables and methods **get()**, **disp()**, **findgrade()**. If total mark is greater than 900 grade A, greater than 700 grade B, greater than 500 grade C else failed. Create a subclass **ProjectStudent** from the Student class with the data members **projid**, **projtitle**, **company**, **projgrade** with methods **getPDetails()**, **dispPDetails()**. Create instance of Project Student and invoke all the methods in main.
   (Do the program without using constructors)
2. Create a class named **EmployeeSample** having the following data members:
   name
   age
   phoneno
   address
   salary
   It also has a method named **printSalary()** which prints the salary of the employee. Another class **Manager** inherits the Employee class. The Manager class have data member **department**. Using constructor assign name, age, phone number, address, salary and department to a manager by making an object of Manager class and print the salary.
3. Create a class **A** with the method **dispmsg()** which prints some message. Create one subclass of A with name **B** and override the dispmsg() function, also create one more subclass from B with name **C** and override the dispmsg() function in C also. In main invoke that function with the help of three class instances.
4. Write a class called **Square** with instance variable **a** and a method **double area()** that finds the area(a2) of the square. Create a class Cube which is a subclass of Square and write an overriding method double area() that finds the surface area (6a2) of the cube using super keyword. In the main function, invoke the function using the two class instances.

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5. Create a class named **Account** with following members

   I. Data Members
      a. accNo
      b. Name
      c. accType
      d. accBalance

   II. Function members
      a. Constructor to accept all values
      b. deposit(int amt) accepting amount and add with the accBalance
      c. withdraw(int amt) accepting amount to subtract from the accBalance
      d. checkBalance() to return the present accBalance

   Inherit classes such as **SavingsAccount** and **PrivilegedAccount** from the class Account. The rules of these account types are as follows:-

   I. SavingsAccount
      a. Deposit – can deposit upto a maximum of 50000 only in one transaction.
      b. Withdraw – a minimum balance of 1000 should be there at any time.

   II. PrivilegedAccount
      a. Deposit – can deposit any amount
      b. Withdraw – can take an overdraft of maximum 5000.ie, account balance go upto - 5000.

   Write a main() function. Create objects of SavingsAccount and PrivilegedAccount, assign values to instance variables using constructor and invoke the methods.

6. Create an abstract class **Vehicle** with one abstract function **color()**. Create two sub classes **Car** and **Bus** from the Vehicle class. Invoke the function through the instance of Car and Bus. Also use the abstract class reference that invokes that function in main.
7. Create an abstract class **Animals** with two abstract methods **eat()** and **makeNoise()**. Create a class **Cats** where makeNoise() method is overridden which prints "Cats meow" and a class **Dogs** where makeNoise() method is overridden which prints "Dogs bark", both inheriting the class Animals. Now create an object for each of the subclasses and call their respective methods.